User Manual for

Micro-A748

v2.0

www.microembedded.in

Table of Contents.

1.0 Introduction.

1.1 Overview

Today we see a lot of new and amazing gadgets in the market. We see high end mobile phones, tablet PC, Netbook PC and state of the art electronic devices. It will amaze us that all these devices are using processors which are having ARM Microprocessor inside it. ARM microprocessor technology is the most efficient technology available today.

Micro-A748 Microcontroller Kit is specifically designed keeping in mind the needs of students, to understand and learn the architecture of the ARM 7 microprocessor.

The Micro-A748 kit has interfaced the ARM7 core based SOC (system on chip), to various peripherals on the board. Devices like LED's and LCD, I2C interface based memory devices, RTC, SD/MMC Card interface, Matrix Keypad, ADC, DAC, graphical LCD, Stepper motor, etc are also interfaced on the board.

ARM Controller.

The ARM Processor is a RISC(Reduced Instruction Set Computer) machine. The RISC instruction set is primarily known for smaller number of instructions and higher throughput at a lower power.

The ARM Architecture is characterized by the following;

1. A load-store architecture.
2. Fixed length 32 bit instructions in ARM Mode and 16 bit instructions in THUMB Mode.
3. 3 address instruction format.

There are several modes of operation. In each mode there are 16 registers that are accessible to the user programs. A Current Program Status Register (CPSR) is also available which contains all the mode bits and the interrupt bit and the condition bits. The modes allow the user program restricted access to certain resources of the processor.

Following are the modes in the ARM Processor.

- User Mode : This is the main operating mode for the processor. Program running in this mode can achieve isolation and protection for the resources.
- Fast Interrupt Processing Mode (FIQ): This mode is entered when the Fast interrupt is received. This is called a fast interrupt as it has a dedicated vector address from where the program can immediately execute without any latency.
- Normal Interrupt Processing mode (IRQ): Interrupt from other sources will make the processor enter this mode.
- Software Interrupt Mode: when the processor encounters a Software Interrupt Instruction this mode is entered. Software Interrupts are a standard way to enter the privileged mode from User Mode.
- Undefined Instruction mode: when a processor attempts to execute a instruction which neither its main core or the co-processor can execute then this mode is entered.
- System mode: Privileged operating system task can be run from this mode.
- Abort Mode: this mode is entered when a data fault is occurred.

<u>The 3 Stage Pipeline.</u>

There is a 3 stage pipeline in the ARM 7 processor.

1. Fetch : Fetch the instruction from memory.
2. Decode : Decode the instruction .
3. Execute.: Execute the instruction.

Due to the 3 stage pipeline the ARM processor is able to effectively execute 1 instruction for every clock cycle. Thus it has a very high throughput and performance.
The ARM Processor supports the following Data Types.
8 Bit signed and unsigned byte.
16 Bit signed and unsigned Half word aligned on 2 byte boundary.
32 bit signed and unsigned Word aligned on 4 byte boundary.
The ARM Processor supports both the BIG Endian and Little Endian Memory alignment.

1.2 Features

The Micro-A748 microcontroller board has been specifically designed keeping in mind the needs of students for learning the ARM architecture. The board gives a complete overview for interfacing various peripheral devices which are used in the industry and consumer devices alike. A hands-on with the board will develop in the student the experience to design and implement various devices and products based on the ARM Microcontroller.

Following are the features of the Micro-A748 Microcontroller board.

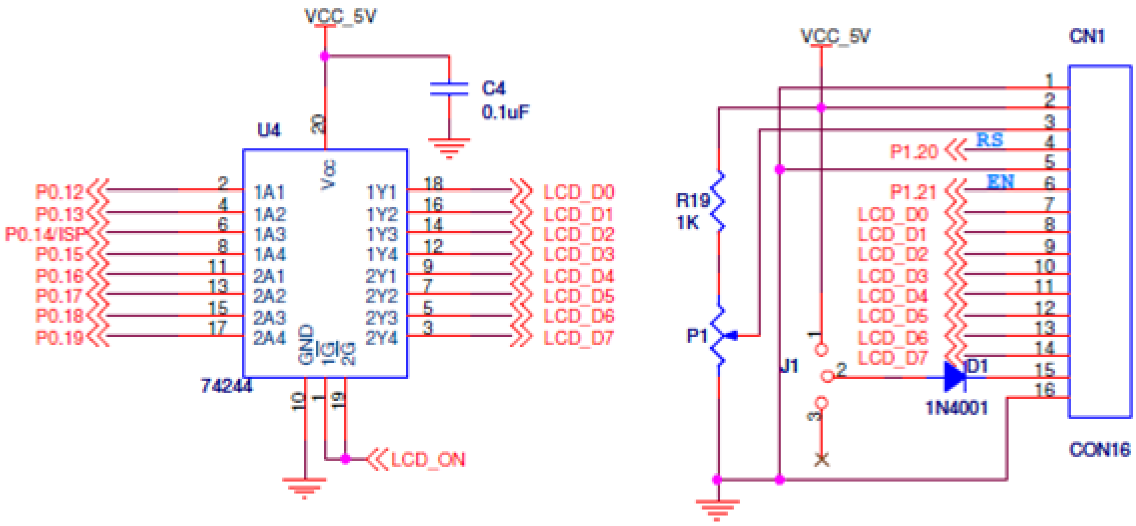
- LPC21xx microcontroller from NXP running at maximum 60 MHz.
- 5 Volt Operation.
- Two Uarts with RS232 drivers.
- 16x2 character LCD module.
- 16 Key Matrix Keypad.
- 8 general purpose LED’s.
- 2 multiplexed Seven Segment display.
- RTC with I2C interface with power backup.
- EEPROM with I2C interface.
- SD/MMC card interface on SPI interface.
- 10 bit on Chip ADC interfaced to External voltage source.
- 10 Bit DAC.
- Stepper Motor and DC Motor Driver (L293D).
- Relay and Buzzer.
- Optional 128X64 graphical LCD Display module.
- Peripheral selection using Jumpers.

1.3 Micro-A748: Board Photo

2.0 Hardware: Functional Description and Interfacing.

The Micro-A748 microcontroller board has been specifically designed keeping in mind the needs of students for learning the ARM architecture. The board gives a complete overview for interfacing various peripheral devices which are used in the industry and consumer devices alike. A hands-on with the board will develop in the student the experience to design and implement various devices and products based on the ARM Microcontroller.

2.1 LCD.



2.1.1 Character LCD.

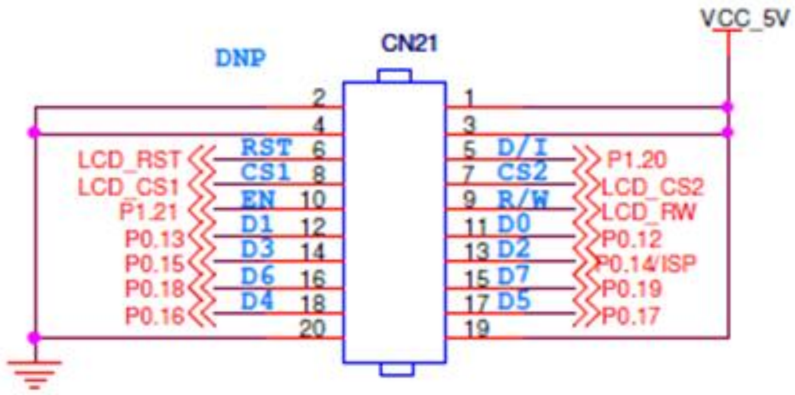
A 16x2 Character LCD module (CN1) is interfaced on the Micro-A748 board. The LCD is write only ie we cannot read from the LCD. Contrast control is adjusted using potentiometer P3.

Put the switch SW21 in position 1- 2 to use the LCD.

LCD interfacing details.

Device	Pin Details
LCD D0	P0.12
LCD D1	P0.13
LCD D2	P0.14
LCD D3	P0.15
LCD D4	P0.16
LCD D5	P0.17
LCD D6	P0.18
LCD D7	P0.19
LCD RS	P1.20
LCD EN	P1.21
Peripheral Selection	
SW21	towards LCD_ON

2.1.2 Graphic LCD.



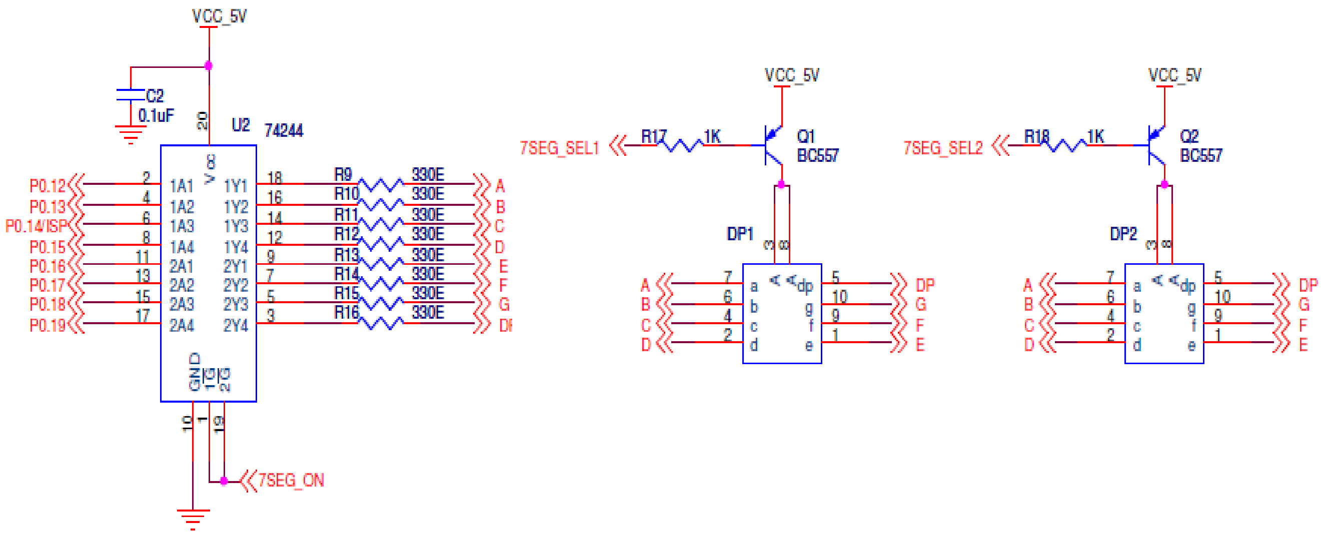
A 128x64 Graphic LCD module is interfaced on the Micro-A748 board. We can read and write data to the LCD module. Contrast control for the LCD is done by using the potentiometer P1.

Put the switch SW21 in position 1- 2 to use the LCD.

Graphics LCD interfacing details.

Device	Pin Details
GLCD D0	P0.12
GLCD D1	P0.13
GLCD D2	P0.14
GLCD D3	P0.15
GLCD D4	P0.16
GLCD D5	P0.17
GLCD D6	P0.18
GLCD D7	P0.19
GLCD EN	P1.20
GLCD DI	P1.21
GLCD CS1	P1.22
GLCD CS2	P1.23
GLCD RST	P1.24
GLCD R/W	P1.25
Peripheral Selection	
SW21	towards LCD_ON

2.2 Seven Segment Display.



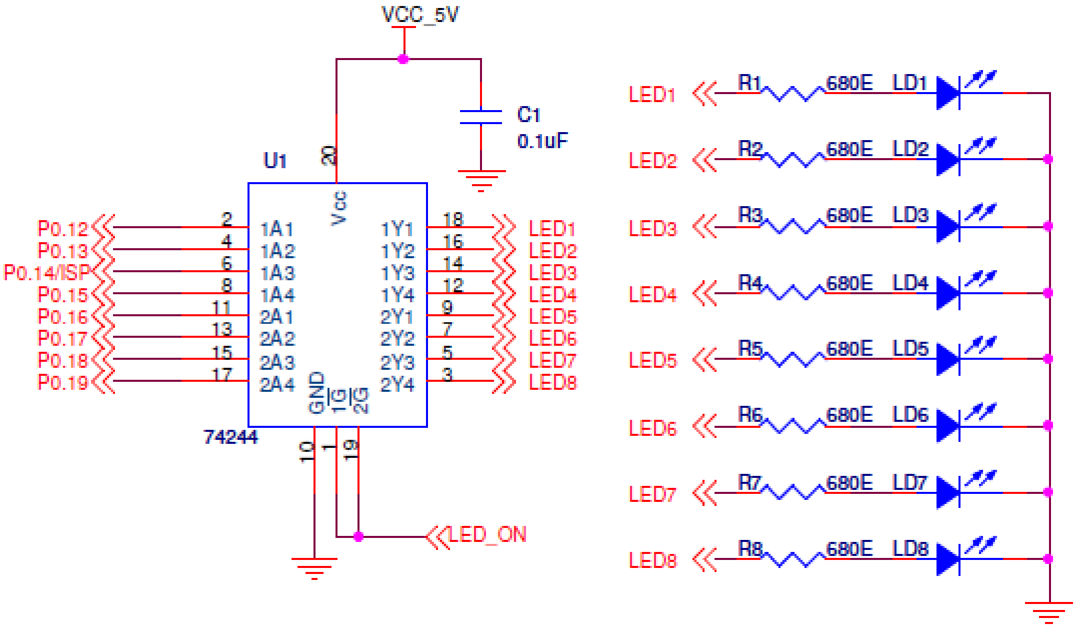
Common Anode 7 Segment Display modules (DP1 and DP2) are interfaced on the Micro-A748 board. In order to make the segment ON Logic “0” (Low) has to be applied to the Port Pin.

In order to select the device for operation put switch SW20 in position 2-3 and switch SW21 in position 2-3.

Seven Segment interfacing details

Device	Pin Details
Segment A	P0.12
Segment B	P0.13
Segment C	P0.14
Segment D	P0.15
Segment E	P0.16
Segment F	P0.17
Segment G	P0.18
Segment DP	P0.19
SS Select1	P1.21
SS Select2	P1.20
Peripheral Selection	
SW 20	towards 7SEG_ON
SW21	away from LCD_ON

2.3 Light Emitting Diodes (LED).



8 general purpose LED's are provided on the Micro-A748 Board in common cathode configuration.LD1 thru LD8 are interface to Port Pins P0.12 thru P0.19.

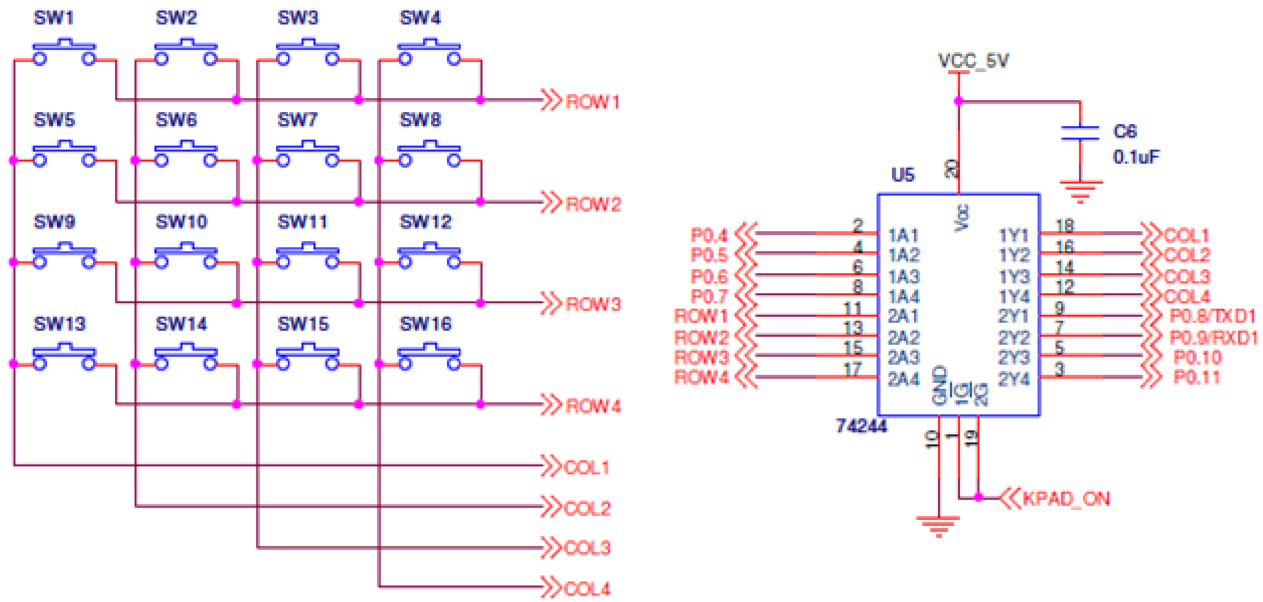
In order to make the LED ON we have to give Logic "0"(LOW).

For this device to operate correctly connect the switch SW20 between 1-2 and SW21 between 2-3.

LED interfacing details.

Device	Pin Details
LED 0	P0.12
LED 1	P0.13
LED 2	P0.14
LED 3	P0.15
LED 4	P0.16
LED 5	P0.17
LED 6	P0.18
LED 7	P0.19
Peripheral Selection	
SW 20	towards LED_ON
SW21	away from LCD_ON

2.4 Keypad.



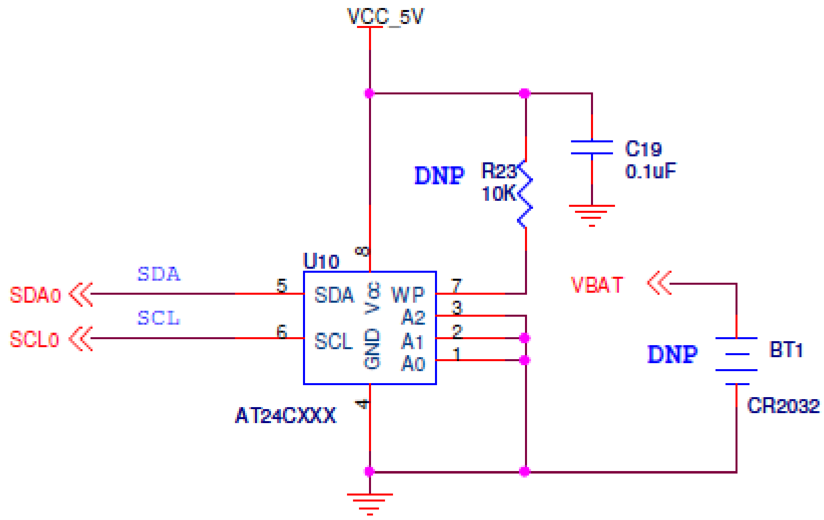
A 16 key Keypad in a 4x4 matrix circuit is provided on the Micro-A748 board. The Keypad is interfaced to the GPIO pins of the processor.

In order to operate this device the connect switch SW22 in position 1-2.

Keypad interfacing details.

Device	Pin Details
KEYPAD COL 0	P0.4
KEYPAD COL 1	P0.5
KEYPAD COL 2	P0.6
KEYPAD COL 3	P0.7
KEYPAD ROW 0	P0.8
KEYPAD ROW 1	P0.9
KEYPAD ROW 2	P0.10
KEYPAD ROW 3	P0.11
Peripheral Selection	
SW22	towards KPAD_ON

2.5 I2C Interface (EEPROM and RTC).



The I2C interface from the processor has been brought out on the Micro-A748 board. The Devices such as EEPROM (AT24Cxx) and RTC (DS1307) with I2C interface have been integrated on the board.

The Port pins have to be configured to work as I2C clock and Data lines. No jumper settings are needed to use this interface.

I2C interfacing details.

Device	Pin Details
I2C SDA	SCL0 (P0.3)
I2C SCL	SDA0 (P0.2)

2.6 SPI Interface.

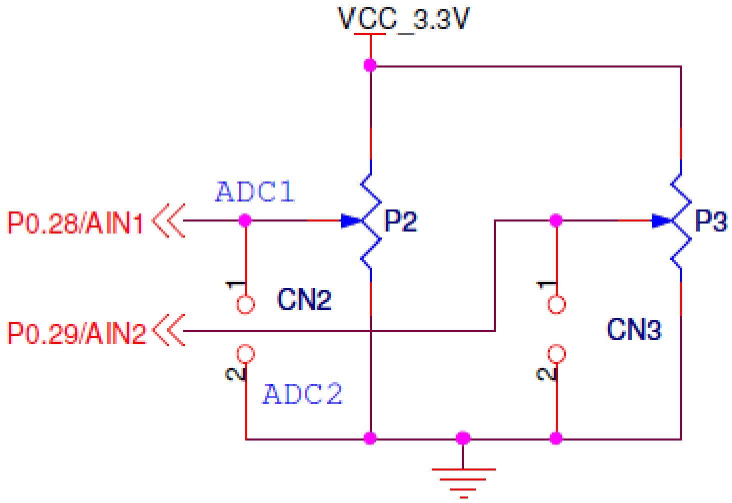
The SPI interface from the processor is available on the Micro-A748 board. Devices such as SPI based SD/MMC Card can be interfaced on the board using the SPI.

The port pins from the processor have to be configured as SPI. No jumper settings are needed to use the SPI interface.

SPI interfacing details.

Device	Pin Details
SPI SCK	P0.4
SPI MISO	P0.5
SPI MOSI	P0.6
SPI SSEL	P0.7

2.7 Analog to Digital Converter (ADC).

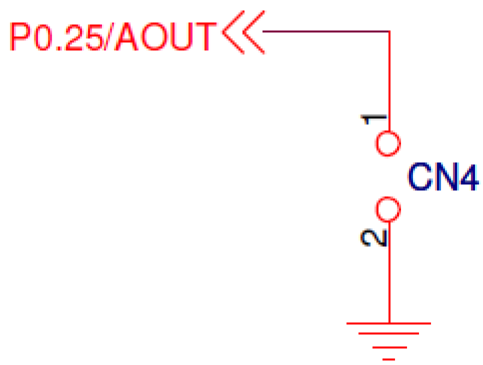


The processor on the Micro-A748 contains a 10 bit ADC with 7 channels. The board has provided the user with 2 channels. Analog voltage is provided using potentiometers P2 and P3. ADC0 peripheral is connected to this input. The Analog voltage can be measured at connector CN2 and CN3 respectively.

ADC interfacing details.

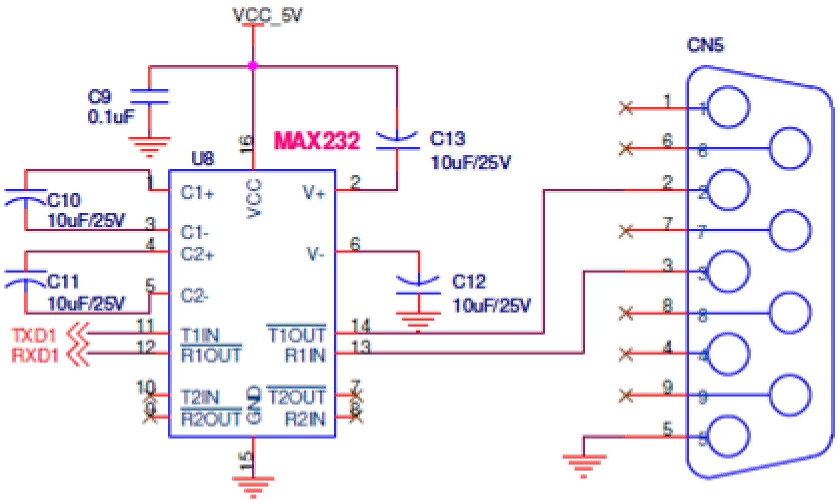
Device	Pin Details
ADC 2	P0.28
ADC 1	P0.29

2.8 Digital to Analog Convertor (DAC).



The Processor on Micro-A748 contains a 10 bit Digital to Analog Convertor. The Output of this DAC is brought out on pin P0.25 on the board at connector CN4.

2.9 Serial Communication using UART.



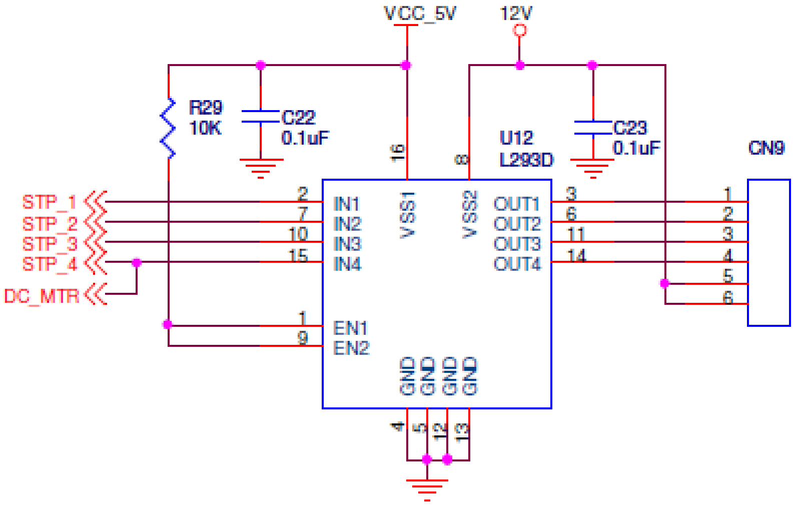
Two UART interfaces from the processor are available for the user on the Micro-A748 board. UART0 is connected to USB to Serial and UART1 is connected to COM2.

In order to use UART1 switch SW23 has to be connected between 1-2.

UART interfacing details.

Device	**Pin Details**
UART0 RXD	P0.1
UART0 TXD	P0.0
UART1 TXD	P0.8
UART01TXD	P0.9

2.10 Stepper Motor and DC Motor.



A unipolar stepper motor driver (L293D) is provided on the Micro-A748 board for the user. The user can connect the stepper motor to connector **CN9**.

In order to use the stepper motor put switch SW26 in position 1- 2 and to use the DC motor connect the switch SW26 in position 2-3.

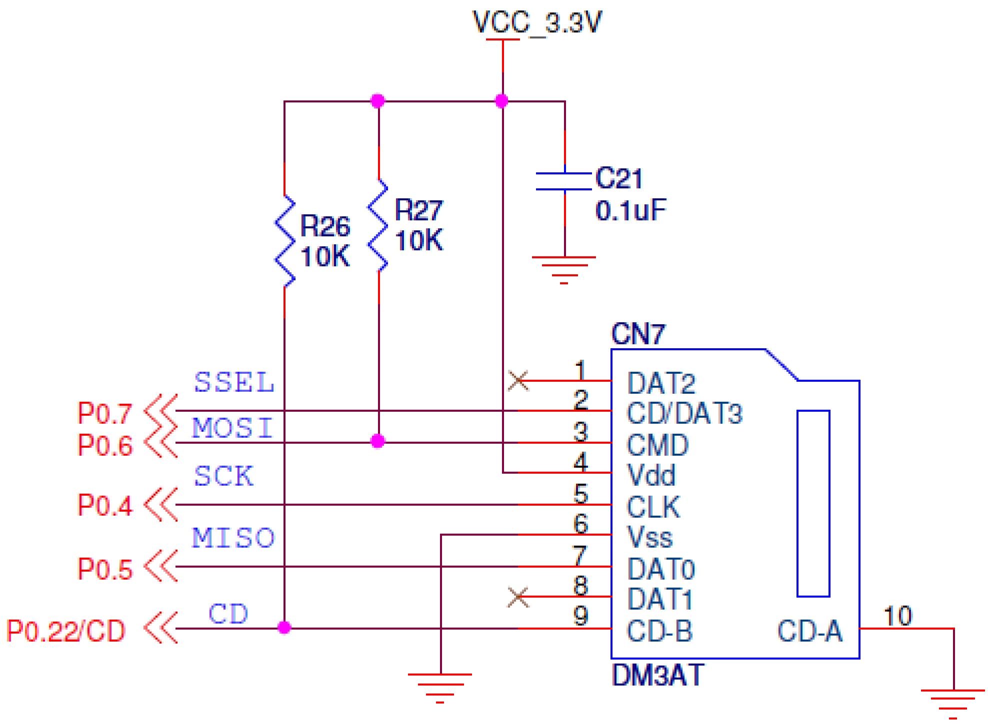
Stepper Motor interfacing details.

Device	Pin Details	CN9
Stepper Motor 1	P1.19	CN 9.1
Stepper Motor 2	P1.18	CN 9.2
Stepper Motor 3	P1.17	CN 9.3
Stepper Motor 4	P1.16	CN 9.4
Peripheral Selection		
SW 26	towards STP_ON	

DC motor interfacing details.

Device	Pin Details	CN9
DC Motor	P0.21	CN 9.4
Peripheral Selection		
SW 26	towards DC_ON	

2.11 SD/MMC Card.

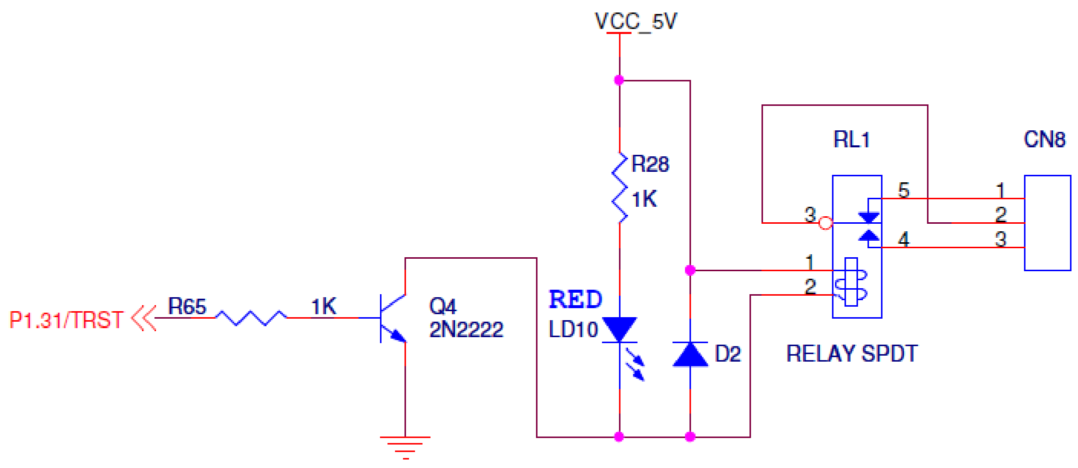


The SD card is interfaced on the Micro-A748 board over the SPI interface.

SD Card interfacing details.

Device	Pin Details
SPI SCK	P0.4 (CLK)
SPI MISO	P0.5 (DAT 0)
SPI MOSI	P0.6 (CMD)
SPI SSEL	P0.7 (DAT 3)
SD CD	P0.22 (Card Detect)

2.12 Relay and Buzzer.

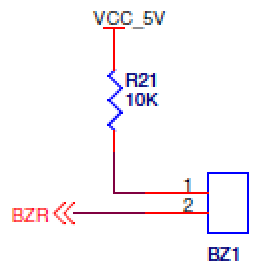


A general purpose relay is connected on the Micro-A748 board for the user to interface external high voltage devices to main or AC supply. The external devices /AC supply can be connected on connector CN8.

Relay interfacing details.

Device	Pin Details
Relay	P1.31

Buzzer:



A general purpose buzzer is connected on the Micro-A748 board for indication purpose.

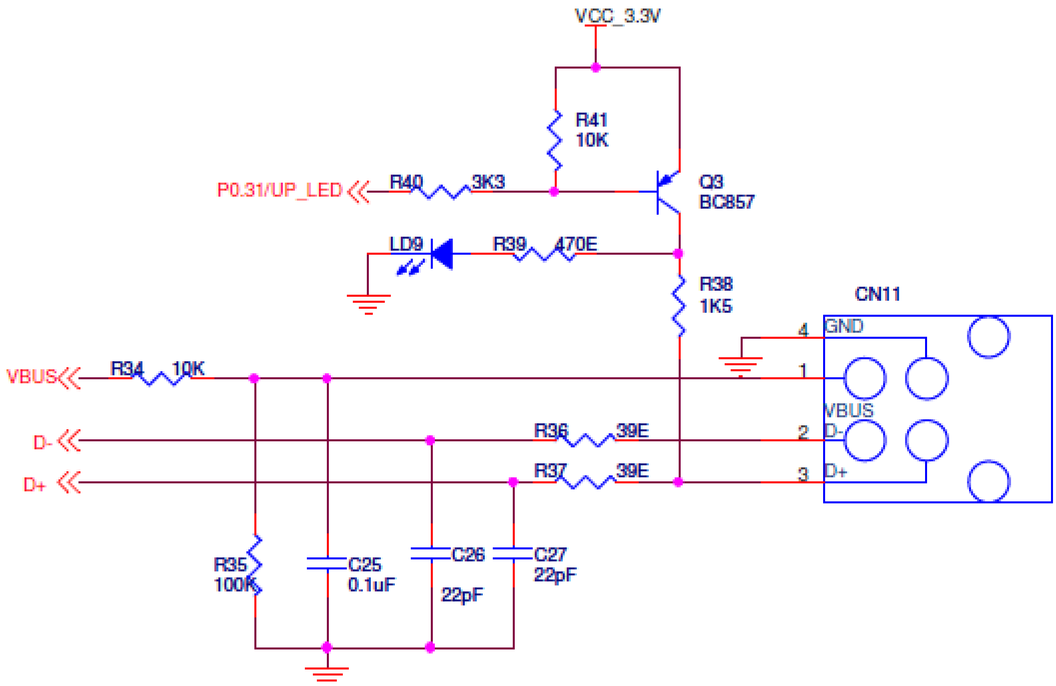
To sound the buzzer ,give logic ‘0’ to the pin P0.6.

To connect the buzzer connect switch SW22 in position 2-3.

Buzzer interfacing details.

Device	Pin Details
BUZZER	P0.6
Peripheral Selection	
SW22	towards DIP_ON

2.13 USB Device Port.



The processor on the Micro-A748 (LPC2148) has a on chip USB device controller and driver interface.

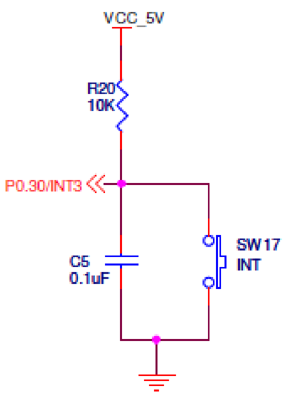
The Micro-A748 has connected this interface to a USB device connector (USB-B type connector).

USB Device interfacing details.

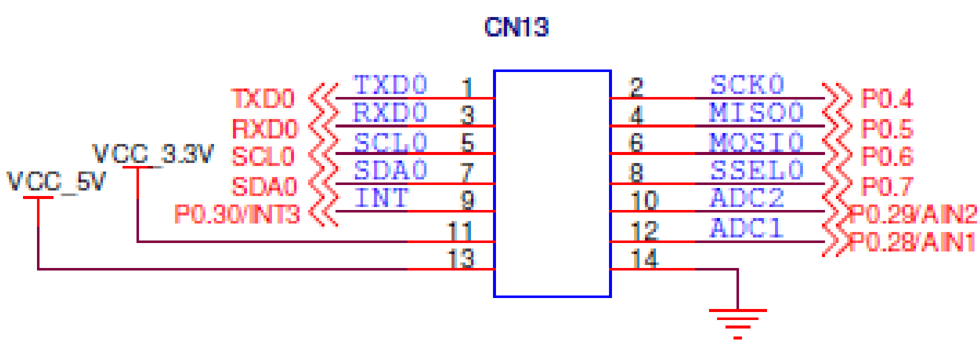
USB Device Interfacing Details	
Device Pins	Port Pins
VBUS	VBUS (Processor Pin 58)
DP	P0.26 (D+)
DM	P0.27 (D-)
LED	P0.31 (UP_LED)

2.14 External Interrupt Switch

The External Interrupt is generated using a switch SW17 connected to Pin P0.30 (INT3).



2.15 Port Extensions.



CN13 interfacing details.

CN13	Pin Details
CN13.1	P0.0
CN13.2	P0.4
CN13.3	P0.1
CN13.4	P0.5
CN13.5	P0.2
CN13.6	P0.6
CN13.7	P0.3
CN13.8	P0.7
CN13.9	P0.30
CN13.10	P0.29
CN13.11	3.3V
CN13.12	P0.28
CN13.13	5V
CN13.14	GND

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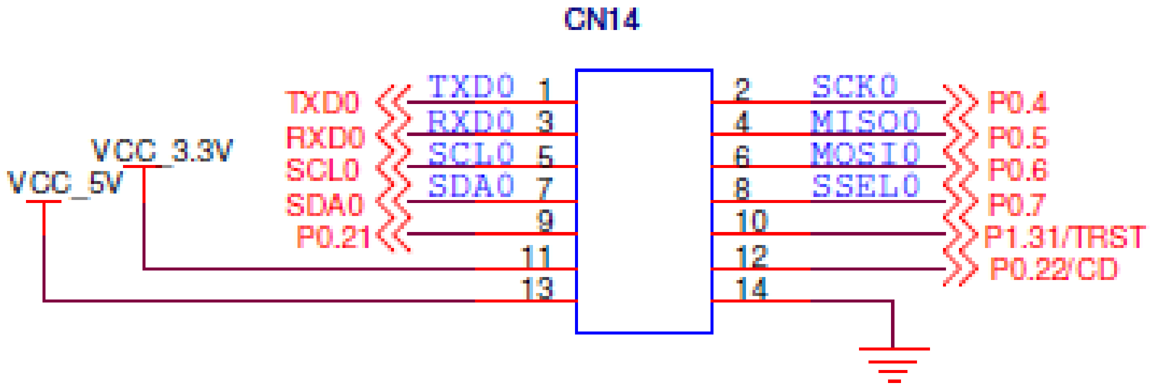
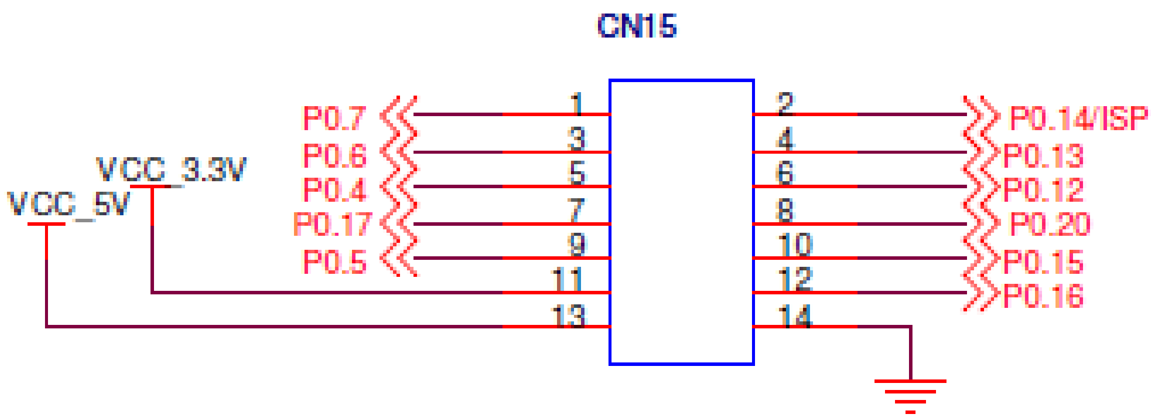


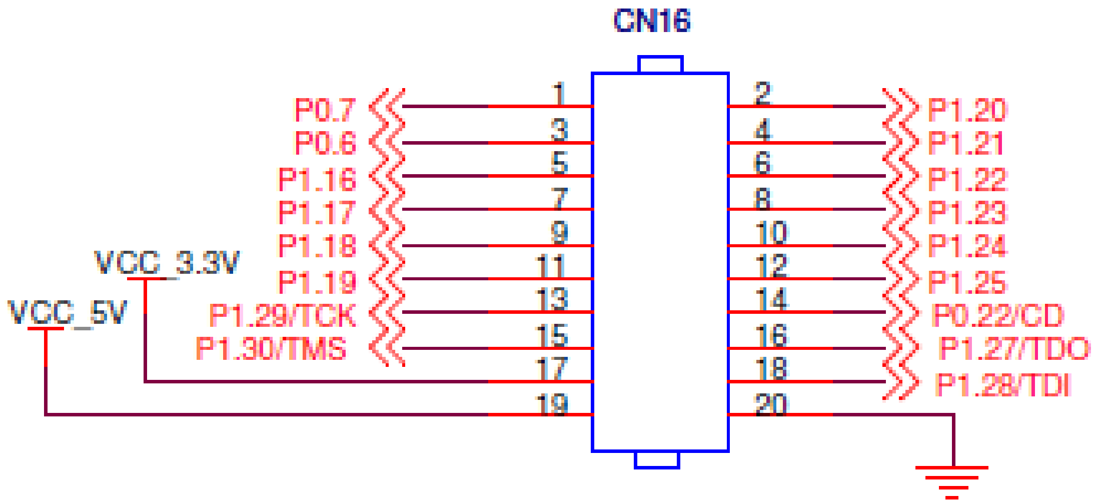
CN14 interfacing details.

CN14	Pin Details
CN14.1	P0.0
CN14.2	P0.4
CN14.3	P0.1
CN14.4	P0.5
CN14.5	P0.2
CN14.6	P0.6
CN14.7	P0.3
CN14.8	P0.7
CN14.9	P1.18
CN14.10	P1.31
CN14.11	3.3V
CN14.12	P0.22
CN14.13	5V
CN14.14	GND



CN15 interfacing details.

CN15	Pin Details
CN15.1	P0.7
CN15.2	P0.14
CN15.3	P0.6
CN15.4	P0.13
CN15.5	P0.4
CN15.6	P0.12
CN15.7	P0.17
CN15.8	P0.20
CN15.9	P0.5
CN15.10	P0.15
CN15.11	3.3 V
CN15.12	P0.16
CN15.13	5V
CN15.14	GND



CN16 interfacing details.

CN16	Pin Details
CN16.1	P0.7
CN16.2	P1.20
CN16.3	P0.6
CN16.4	P1.21
CN16.5	P1.19
CN16.6	P1.22
CN16.7	P1.18
CN16.8	P1.23
CN16.9	P1.17
CN16.10	P1.24
CN16.11	P1.16
CN16.12	P1.25
CN16.13	P1.29
CN16.14	P0.22
CN16.15	P1.30
CN16.16	P1.27
CN16.17	3.3V
CN16.18	P1.28
CN16.19	5V
CN16.20	GND

3.0 Software : Functional Description and Interfacing.

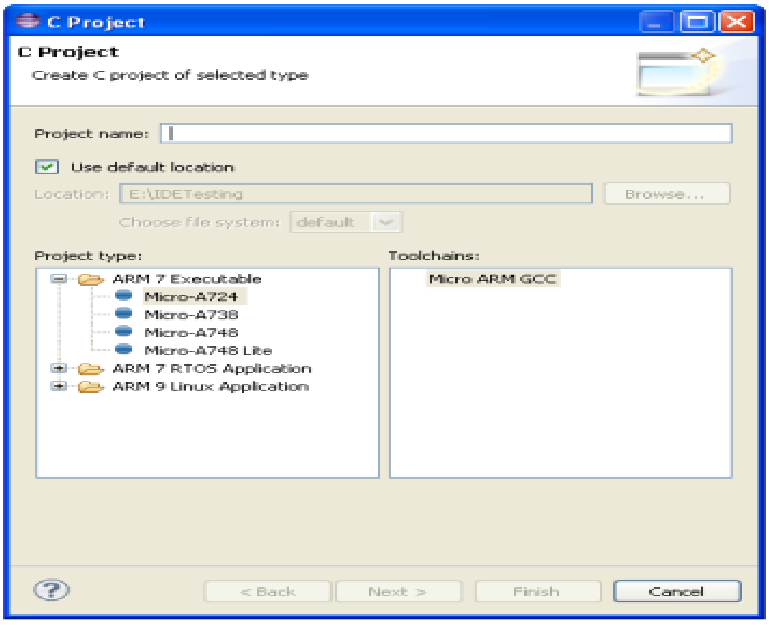
3.1 PC Connection Setup.

The Micro-A748 Board connects to the PC via the USB to Serial port. The Hex files generated by the IDE can be downloaded into the flash memory using the ISP feature on the microcontroller via the serial port. The Micro-A748 Board development kit has a serial port cable included in the package. Connect the female side of the connector to the PC and the male side to the Micro-A748 Board.

3.3 µC-Igniter Programming IDE.

Creating a new project.

- Go to the Project Tab.
- Click on New --->C Project.



- Enter the project name.
- Choose the appropriate board name.

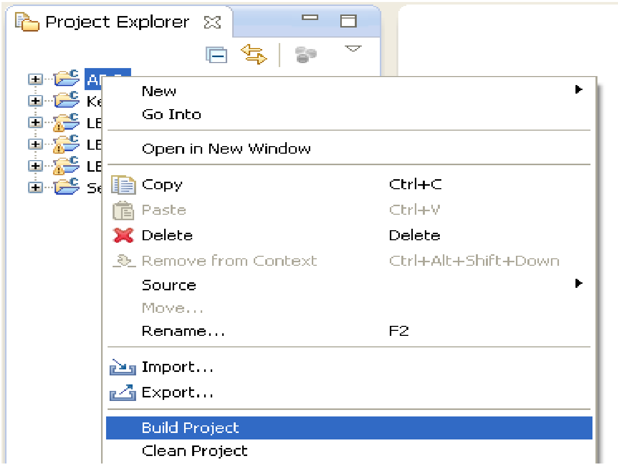
ARM 7 Executable for Standalone application for ARM7.

ARM 7 RTOS Application for RTOS applications on ARM7.

ARM9 Linux Application for Linux Programs on ARM9.

- Click Finish to complete.

Compiling the project

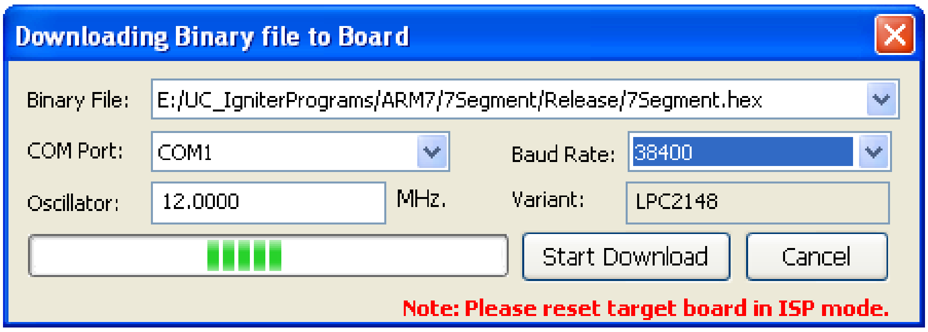


- Right Click on the Project and select **Build Project.** After successful Build a **.hex** file will be generated in the Release folder.

Downloading the hex file on the board.

ARM7 based processors.

- Right Click on the Project and Click on **Download on Target.**

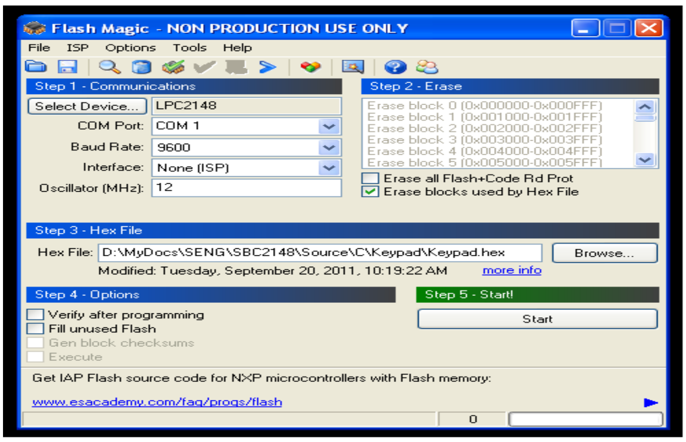


Put the RUN/ISP switch on ISP Mode and press reset Switch.

- Click on Start Download.
- The hex file will get downloaded successfully.

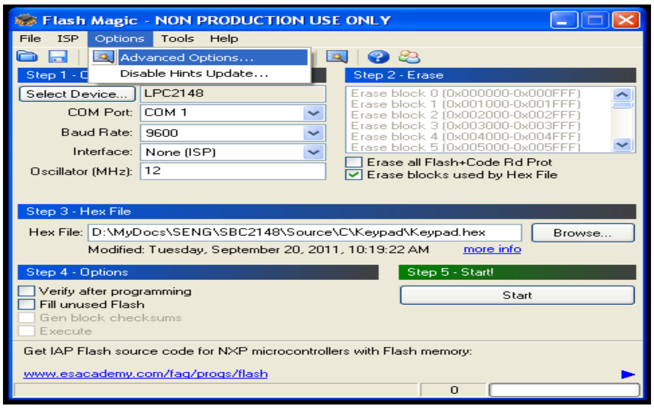
3.4 Executable Flashing Tool (FlashMagic).

Flash Magic Software is used to program the microcontroller on the Micro-A748 Board.



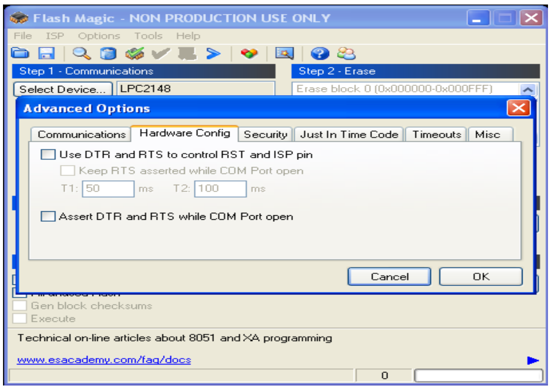
The following setting are to be done to correctly use Flash Magic with Micro-A748 board.

- Select Device as LPC2148.
- Select the correct COM Port. (Port on PC over which serial cable is connected).
- Set baud Rate to 9600.
- Interface: None.
- Oscillator: 12 MHz.
- **Check the box: Erase Blocks Used By hex File.**
  Note: Do not select: Erase all Flash + Code Rd Prot.



Hardware Settings for Flash Magic.

- Select Options tab.



- Select Hardware Config and uncheck USE DTR and RTS to control RST and ISP Pin.

1. **Flashing the Micro-A748 Board**.

Following are the steps to be done while programming the HEX file to the microcontroller Flash Memory.

Power "ON" the board using the ON/OFF Switch.

- Set the RUN/ISP switch to ISP Mode.
- Start Flash Magic and select the Hex file to be downloaded.
- Press the Reset button of the board.
- Press Start button on Flash Magic.
- Check if the program is downloaded.
- Set the RUN/ISP Switch to RUN Mode.
- Press the Reset Button on the board to start the program in the microcontroller.